

Twinkle Kamdar

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EDUCATION

Carnegie Mellon University

Masters of Science in Information Security | GPA: 3.6/4

Pittsburgh, PA

December 2026

Pandit Deendayal Energy University

B.Tech. in Computer Science Engineering | CGPA: 9.44/10

Gandhinagar, India

May 2025

EXPERIENCE

Floatbot.AI

Remote · Milpitas, CA

Security Analyst Intern

May 2026 – Present

- Conducted security design reviews for Floatbot's conversational AI infrastructure, threat modeled distributed agent access boundaries, API exposure, and data flow risks across GCP US, Azure UAE, and GCP India CDE environments; produced attack path models mapping lateral movement risks.
- Built PCI DSS v4.0.1 compliance posture management framework, controls tracker, evidence mapping, scoping documents, and System Security Plan defining CDE perimeter, data flows, and access invariants across distributed cloud environments.

Silver Touch Technologies

Ahmedabad, India

SOC Intern

May 2024 – July 2024

- Performed VAPT and security assessments for enterprise clients, triaged 50+ CVEs using CVSS/CWE; conducted attack path modeling to identify non-obvious lateral movement and indirect access risks.
- Monitored SIEM for alert triage; distilled access intelligence from internal systems to identify anomalies and prioritize high-risk remediation; maintained NIST CSF and SOC 2 aligned security documentation.

PROJECTS AND RESEARCH

IRIS: LLM Agent Security Monitor | GitHub | [iris-hell99.streamlit.app](#)

March 2026 – Present

- Built 5-layer agentic identity security posture management framework, 894 tool calls monitored, 90 blocked, 14 cross-agent collusion patterns detected; OPA declarative policy engine enforcing security invariants.
- Intent-Action Divergence engine (llama-3.3-70b) catches indirect prompt injection bypassing LLM safety training, 93.1% precision, 83.1% F1 on 50 novel adversarial scenarios; CUSUM drift detection, MITRE ATLAS TTP mapping; pip install iris-security.

AEGIS: Autonomous Cybersecurity Intelligence System | GitHub

Dec 2025 – March 2026

- 3-layer ensemble ML IDS/IPS, 99% accuracy across 2.5M+ flows, autonomous blocking in <5ms; post-quantum crypto (Dilithium3 + Kyber768), SHA-256 audit ledger, security invariant enforcement via self-healing watchdog.
- Kubernetes Zero Trust with strict RBAC and network policy enforcement; NIST CSF IR automation; integrated Mininet SDN for simulation, hardening critical service attack surface by eliminating unnecessary access.

ZeroSeg: AI-Powered Access Risk Intelligence & Microsegmentation | GitHub

Jan 2026 – May 2026

- XGBoost (95.31% accuracy) + DBSCAN (29 segments) on 51,628 flows, distills access intelligence to detect lateral movement TTPs (MITRE ATT&CK TA0007/TA0008) and prioritize high-risk remediation.
- Deny-by-default OpenFlow 1.3 mandatory access control, 100% cross-segment block, 0 false positives, <1ms latency; real-time XGBoost risk score dashboard.

SKILLS

Programming Languages: Python | C | C++ | Go | Bash | Java | SQL | JavaScript

Security Engineering: Threat Modeling | Attack Path Modeling | Security Posture Management | Security Invariant Enforcement | PAM | Agentic Identity Security | IAM | Zero Trust | Least Privilege | Access Intelligence

ML & Detection: XGBoost | DBSCAN | MITRE ATLAS | Behavioral Analysis | Intent-Action Divergence

Tools: Splunk | Suricata | Zeek | Falco | Security Onion | Wireshark | Burp Suite | Metasploit | LangGraph | LangChain | FastAPI

Cloud & Infrastructure: AWS | GCP | Azure | Kubernetes | Docker | Terraform | OpenFlow | GNS3 | Mininet

Frameworks: NIST CSF | PCI DSS v4.0.1 | ISO 27001:2022 | HIPAA | GDPR | OWASP | CVSS/CWE | SOC 2 | Incident Response | Detection Engineering

Certifications: IBM Threat Intelligence & Hunting | Cisco Introduction to Cybersecurity | PicoCTF: CMU